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EXPERIMENT NO. 2

STUDY AND EXECUTION OF BASIC NETWORKING COMMANDS

IN LINUX/NETKIT

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AIM:

To understand and execute basic networking commands used for configuration, monito ring, and troubleshooting.

OBJECTIVES

- To study commonly used networking commands in Linux/Netkit.
- To configure and display network interface parameters.
- To test network connectivity between hosts.
- To monitor network status and routing information.
- To perform basic network troubleshooting using command-line tools**.**

THEORY

Linux provides command line utilities for interface configuration, connectivity testing, r oute inspection, socket monitoring and troubleshooting.

Commands such as ip, ifconfig, ping, netstat, traceroute, ss, ethtool, iwconfig, curl and wget are useful in practical networking tasks.

SOFTWARE / MATERIALS REQUIRED

- Ubuntu/Linux Operating System or Netkit
- Terminal Access
- Internet/Network Connectivity

- **IMPORTANT COMMANDS / PROGRAM STEPS**

```
ip address ip route

ifconfig  ping

example.com netstat -a

traceroute example.com ss

-t -a

ethtool eth0 iwconfig

curl https://example.com

wget

https://example.com/file.

zip
```

PROCEDURE

1. Open the Linux terminal.
2. Execute each networking command one by one.
3. Observe and note interface configuration details.
4. Test connectivity using ping.
5. Check routing information and active connections.
6. Record the observations in the observation table.

- **OBSERVATION / COMPARISON TABLES**

Command	Purpose	Observation
ip address	Display IP configuration	Showed IP addresses, netmasks, and link states for all network interfaces
ifconfig	Display interface details	Checked interface details (ens33, lo), MAC addresses, and tested bringing interfaces up/down and changing IPs
ping	Test connectivity	Tested local stack and network connectivity with 0% packet loss, verifying custom counts, intervals, and payload sizes
netstat	Show network statistics	Displayed active network connections, listening sockets, routing tables, and protocol filters
traceroute	Trace route path	Traced network paths to reachable hosts and tested responses for unreachable subnets
ss	Show socket information	Displayed detailed socket statistics and connection states
ethtool	Show Ethernet details	Showed Ethernet device parameters and link status
iwconfig	Show wireless details	Displayed wireless interface parameters and link quality

RESULT:

1. Switching to Root & Checking Interfaces (sudo su + ifconfig)

Elevates to the root shell, then runs ifconfig to list all network interfaces. Shows ens33 (the Ethernet interface) with IP 192.168.152.130, MAC address, and RX/TX packet stats, plus the lo loopback interface (127.0.0.1).

```
siesgst@siesgst-VMware-Virtual-Platform:~$ sudo su
[sudo] password for siesgst:
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.152.130  netmask 255.255.255.0  broadcast 192.168.152.255
        inet6 fe80::20c:29ff:fe77:33a5  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:77:33:a5  txqueuelen 1000  (Ethernet)
        RX packets 70  bytes 17020 (17.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 115  bytes 14103 (14.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 136  bytes 12096 (12.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 136  bytes 12096 (12.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

2. Testing Loopback with ping 127.0.0.1

Pings the loopback address to verify the local TCP/IP stack is working without touching the network. Ten replies come back instantly (sub-millisecond), then Ctrl+Z suspends the running ping job.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ping 127.0.0.1
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.166 ms
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.081 ms
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.069 ms
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.068 ms
64 bytes from 127.0.0.1: icmp_seq=5 ttl=64 time=0.070 ms
64 bytes from 127.0.0.1: icmp_seq=6 ttl=64 time=0.151 ms
64 bytes from 127.0.0.1: icmp_seq=7 ttl=64 time=0.053 ms
64 bytes from 127.0.0.1: icmp_seq=8 ttl=64 time=0.040 ms
64 bytes from 127.0.0.1: icmp_seq=9 ttl=64 time=0.143 ms
64 bytes from 127.0.0.1: icmp_seq=10 ttl=64 time=0.047 ms
^Z
[1]+  Stopped                 ping 127.0.0.1
```

3. Bringing an Interface Down (ifconfig ens33 down)

Disables the ens33 interface. A follow-up ifconfig confirms only lo remains active, and ifconfig ens33 shows the interface now lacks the UP flag and has no IP assigned.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33 down
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 158  bytes 13922 (13.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 158  bytes 13922 (13.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33
ens33: flags=4098<BROADCAST,MULTICAST>  mtu 1500
        ether 00:0c:29:77:33:a5  txqueuelen 1000  (Ethernet)
        RX packets 94  bytes 18968 (18.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 127  bytes 15109 (15.1 KB)
        TX errors 0  dropped 1 overruns 0  carrier 0  collisions 0
```

4. Bringing an Interface Back Up (ifconfig ens33 up)

Re-enables ens33. The interface regains the UP flag and its original IP address (192.168.152.130) is restored.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33 up
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.152.130  netmask 255.255.255.0  broadcast 192.168.152.255
        inet6 fe80::20c:29ff:fe77:33a5  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:77:33:a5  txqueuelen 1000  (Ethernet)
        RX packets 207  bytes 101145 (101.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 226  bytes 38963 (38.9 KB)
        TX errors 0  dropped 1 overruns 0  carrier 0  collisions 0
```

5. Assigning a New IP/Netmask (ifconfig ens33 10.0.0.8 netmask 255.0.0.0)

Manually changes the interface's IP address and subnet mask. Confirmed by the followup ifconfig ens33, which now shows inet 10.0.0.8 with a Class A netmask.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33 10.0.0.8 netmask 255.0.0.0
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.0.8  netmask 255.0.0.0  broadcast 10.255.255.255
        inet6 fe80::20c:29ff:fe77:33a5  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:77:33:a5  txqueuelen 1000  (Ethernet)
        RX packets 990  bytes 1039884 (1.0 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 435  bytes 63544 (63.5 KB)
        TX errors 0  dropped 1 overruns 0  carrier 0  collisions 0
```

6. Disabling the Interface with the New IP (ifconfig ... down)

Takes ens33 down while it's still configured with the 10.0.0.8 address, shown twice to confirm the interface stays down (no UP flag, no inet line).

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33 10.0.0.8 netmask 255.0.0.0 down
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33
ens33: flags=4098<BROADCAST,MULTICAST>  mtu 1500
        inet 10.0.0.8  netmask 255.0.0.0  broadcast 10.255.255.255
        ether 00:0c:29:77:33:a5  txqueuelen 1000  (Ethernet)
        RX packets 1134  bytes 1057941 (1.0 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 496  bytes 68715 (68.7 KB)
        TX errors 0  dropped 1 overruns 0  carrier 0  collisions 0

root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33 down
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33
ens33: flags=4098<BROADCAST,MULTICAST>  mtu 1500
        ether 00:0c:29:77:33:a5  txqueuelen 1000  (Ethernet)
        RX packets 1134  bytes 1057941 (1.0 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 496  bytes 68715 (68.7 KB)
        TX errors 0  dropped 1 overruns 0  carrier 0  collisions 0
```

7. Restoring Original Network Settings

Reassigns the original IP (192.168.152.130) and netmask, and brings the interface back up in one command — returning the VM to its normal network configuration.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33 192.168.152.130  netmask 255.255.255.0 up
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.152.130  netmask 255.255.255.0  broadcast 192.168.152.255
        ether 00:0c:29:77:33:a5  txqueuelen 1000  (Ethernet)
        RX packets 1150  bytes 1059579 (1.0 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 508  bytes 70113 (70.1 KB)
        TX errors 0  dropped 1 overruns 0  carrier 0  collisions 0
```

8. Adding a Static Route (ip route add)

Adds a route for the 192.168.45.0/24 network via gateway 192.168.152.130. ip route then lists the new entry alongside the existing local subnet route.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route add 192.168.45.0/24 via 192.168.152.130
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route
192.168.45.0/24 via 192.168.152.130 dev ens33
192.168.152.0/24 dev ens33 proto kernel scope link src 192.168.152.130
```

9. Deleting a Static Route (ip route del)

Removes the previously added route. The routing table shown afterward confirms only the default local subnet route remains.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route del 192.168.45.0/24 via 192.168.152.130
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route
192.168.152.0/24 dev ens33 proto kernel scope link src 192.168.152.130
```

10. Adding then Replacing a Route (ip route add → ip route replace)

First re-adds the 192.168.45.0/24 route via .130, then uses replace to change its gateway to 192.168.152.135 without deleting and re-adding — useful for updating a route in one step.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route add 192.168.45.0/24 via 192.168.152.130
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route
192.168.45.0/24 via 192.168.152.130 dev ens33
192.168.152.0/24 dev ens33 proto kernel scope link src 192.168.152.130
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route replace 192.168.45.0/24 via 192.168.152.135
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route
192.168.45.0/24 via 192.168.152.135 dev ens33
192.168.152.0/24 dev ens33 proto kernel scope link src 192.168.152.130
```

11. Cleaning Up the Replaced Route (ip route del)

Deletes the route that was pointing to 192.168.152.135, returning the routing table to its default state.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route del 192.168.45.0/24 via 192.168.152.135
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ip route
192.168.152.0/24 dev ens33 proto kernel scope link src 192.168.152.130
```

12. Viewing Active Connections (netstat)

Lists active UDP connections (all to a DNS server at 192.168.152.2) and active UNIX domain sockets used for local inter-process communication.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 10.0.0.8:50873          192.168.152.2:domain    ESTABLISHED
udp        0      0 10.0.0.8:34853          192.168.152.2:domain    ESTABLISHED
udp        0      0 10.0.0.8:51993          192.168.152.2:domain    ESTABLISHED
udp        0      0 10.0.0.8:44192          192.168.152.2:domain    ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  3      [ ]         STREAM     CONNECTED     48512
unix  3      [ ]         STREAM     CONNECTED     35820
unix  3      [ ]         STREAM     CONNECTED     34636
unix  3      [ ]         STREAM     CONNECTED     25784    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     37918
unix  3      [ ]         STREAM     CONNECTED     36706    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     33788
unix  3      [ ]         DGRAM      CONNECTED     33672
unix  2      [ ]         DGRAM                    32977
unix  3      [ ]         STREAM     CONNECTED     37911    /run/user/1000/at-spi/bus
unix  3      [ ]         STREAM     CONNECTED     35687
unix  3      [ ]         STREAM     CONNECTED     36701    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     36468
unix  3      [ ]         STREAM     CONNECTED     37356    /run/systemd/journal/stdout
unix  3      [ ]         STREAM     CONNECTED     36448
unix  2      [ ]         DGRAM      CONNECTED     26811
unix  2      [ ]         DGRAM      CONNECTED     25731
unix  3      [ ]         STREAM     CONNECTED     37974    /run/user/1000/bus
unix  3      [ ]         STREAM     CONNECTED     36045
unix  2      [ ]         DGRAM      CONNECTED     24649
```

13. Listing All Sockets (netstat -a)

Shows every socket — both listening and established — including TCP services like mysql, nfs, ipp (printing), and sunrpc, plus their IPv6 (tcp6) equivalents.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:telnet          0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:nfs             0.0.0.0:*               LISTEN
tcp        0      0 _localdnsstub:domain    0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:sunrpc          0.0.0.0:*               LISTEN
tcp        0      0 _localdnsproxy:domain   0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:37317           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:58075           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:55865           0.0.0.0:*               LISTEN
tcp        0      0 localhost:mysql         0.0.0.0:*               LISTEN
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:42445           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:42727           0.0.0.0:*               LISTEN
tcp        0      0 localhost:33060         0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:1883            0.0.0.0:*               LISTEN
tcp6       0      0 [::]:telnet             [::]:*                  LISTEN
tcp6       0      0 [::]:nfs                [::]:*                  LISTEN
tcp6       0      0 [::]:sunrpc             [::]:*                  LISTEN
tcp6       0      0 [::]:http               [::]:*                  LISTEN
tcp6       0      0 [::]:39585              [::]:*                  LISTEN
tcp6       0      0 [::]:51751              [::]:*                  LISTEN
tcp6       0      0 ip6-localhost:ipp       [::]:*                  LISTEN
tcp6       0      0 [::]:33839              [::]:*                  LISTEN
tcp6       0      0 [::]:60703              [::]:*                  LISTEN
```

14. Listing Only Listening Sockets (netstat -l)

Filters the output to just servers in LISTEN state, making it easier to see which services are waiting for incoming connections.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# netstat -l
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:telnet          0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:nfs             0.0.0.0:*               LISTEN
tcp        0      0 _localdnsstub:domain    0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:sunrpc          0.0.0.0:*               LISTEN
tcp        0      0 _localdnsproxy:domain   0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:37317           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:58075           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:55865           0.0.0.0:*               LISTEN
tcp        0      0 localhost:mysql         0.0.0.0:*               LISTEN
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:42445           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:42727           0.0.0.0:*               LISTEN
tcp        0      0 localhost:33060         0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:1883            0.0.0.0:*               LISTEN
tcp6       0      0 [::]:telnet             [::]:*                  LISTEN
tcp6       0      0 [::]:nfs                [::]:*                  LISTEN
tcp6       0      0 [::]:sunrpc             [::]:*                  LISTEN
tcp6       0      0 [::]:http               [::]:*                  LISTEN
tcp6       0      0 [::]:39585              [::]:*                  LISTEN
tcp6       0      0 [::]:51751              [::]:*                  LISTEN
```

15. Filtering by Protocol — TCP Only (netstat -t)

Restricts output to TCP connections only, hiding UDP and UNIX socket entries.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# netstat -t
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
```

16. Filtering by Protocol — UDP Only (netstat -u)

Shows only UDP connections — in this case, four established UDP sessions to the DNS resolver at 192.168.152.2.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# netstat -u
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 10.0.0.8:50873          192.168.152.2:domain    ESTABLISHED
udp        0      0 10.0.0.8:34853          192.168.152.2:domain    ESTABLISHED
udp        0      0 10.0.0.8:51993          192.168.152.2:domain    ESTABLISHED
udp        0      0 10.0.0.8:44192          192.168.152.2:domain    ESTABLISHED
```

17. Viewing the Kernel Routing Table (netstat -r)

Displays the routing table in the classic netstat format (Destination, Gateway, Genmask, Flags, Iface) — an alternative view to ip route.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# netstat -r
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
192.168.152.0   0.0.0.0         255.255.255.0   U         0 0          0 ens33
```

18. Traceroute to a Reachable vs. Unreachable Host (traceroute)

Traces the path to 192.168.152.5 (same subnet, resolves in 1 hop) and to 192.168.151.5 (different, unconfigured subnet → "Network is unreachable"), followed by a ping to confirm the reachable host responds normally.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# traceroute 192.168.152.5
traceroute to 192.168.152.5 (192.168.152.5), 30 hops max, 60 byte packets
 1  siesgst-VMware-Virtual-Platform (192.168.152.130)  3096.543 ms !H  3096.035 ms !H  3096.014 ms !H
root@siesgst-VMware-Virtual-Platform:/home/siesgst# traceroute 192.168.151.5
traceroute to 192.168.151.5 (192.168.151.5), 30 hops max, 60 byte packets
connect: Network is unreachable
root@siesgst-VMware-Virtual-Platform:/home/siesgst# traceroute 192.168.152.5
traceroute to 192.168.152.5 (192.168.152.5), 30 hops max, 60 byte packets
 1  siesgst-VMware-Virtual-Platform (192.168.152.130)  3111.796 ms !H  3111.657 ms !H  3111.649 ms !H
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ping 192.168.152.130
PING 192.168.152.130 (192.168.152.130) 56(84) bytes of data.
64 bytes from 192.168.152.130: icmp_seq=1 ttl=64 time=0.038 ms
64 bytes from 192.168.152.130: icmp_seq=2 ttl=64 time=0.065 ms
64 bytes from 192.168.152.130: icmp_seq=3 ttl=64 time=0.060 ms
64 bytes from 192.168.152.130: icmp_seq=4 ttl=64 time=0.054 ms
64 bytes from 192.168.152.130: icmp_seq=5 ttl=64 time=0.056 ms
64 bytes from 192.168.152.130: icmp_seq=6 ttl=64 time=0.082 ms
64 bytes from 192.168.152.130: icmp_seq=7 ttl=64 time=0.067 ms
64 bytes from 192.168.152.130: icmp_seq=8 ttl=64 time=0.056 ms
^C
--- 192.168.152.130 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7191ms
rtt min/avg/max/mdev = 0.038/0.059/0.082/0.011 ms
```

19. Basic Ping Test (ping 192.168.152.130)

A standard ping with 4 replies, ending in summary statistics (packets sent/received, 0% loss, round-trip time min/avg/max).

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ping 192.168.152.130
PING 192.168.152.130 (192.168.152.130) 56(84) bytes of data.
64 bytes from 192.168.152.130: icmp_seq=1 ttl=64 time=0.030 ms
64 bytes from 192.168.152.130: icmp_seq=2 ttl=64 time=0.048 ms
64 bytes from 192.168.152.130: icmp_seq=3 ttl=64 time=0.077 ms
64 bytes from 192.168.152.130: icmp_seq=4 ttl=64 time=0.034 ms

--- 192.168.152.130 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3080ms
rtt min/avg/max/mdev = 0.030/0.047/0.077/0.018 ms
```

20. Ping with a Fixed Count (ping -c 9)

Uses the -c flag to send exactly 9 ICMP packets and then stop automatically, rather than running until interrupted.

```
^Croot@siesgst-VMware-Virtual-Platform:/home/siesgst# ping -c 9 192.168.152.130
PING 192.168.152.130 (192.168.152.130) 56(84) bytes of data.
64 bytes from 192.168.152.130: icmp_seq=1 ttl=64 time=0.028 ms
64 bytes from 192.168.152.130: icmp_seq=2 ttl=64 time=0.076 ms
64 bytes from 192.168.152.130: icmp_seq=3 ttl=64 time=0.054 ms
64 bytes from 192.168.152.130: icmp_seq=4 ttl=64 time=0.074 ms
64 bytes from 192.168.152.130: icmp_seq=5 ttl=64 time=0.061 ms
64 bytes from 192.168.152.130: icmp_seq=6 ttl=64 time=0.080 ms
64 bytes from 192.168.152.130: icmp_seq=7 ttl=64 time=0.042 ms
64 bytes from 192.168.152.130: icmp_seq=8 ttl=64 time=0.071 ms
64 bytes from 192.168.152.130: icmp_seq=9 ttl=64 time=0.296 ms

--- 192.168.152.130 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8217ms
rtt min/avg/max/mdev = 0.028/0.086/0.296/0.075 ms
```

21. Ping with Custom Intervals (ping -i 2 and ping -i 0.02)

Demonstrates the -i flag controlling the time between packets — first a slow 2-second interval (manually stopped with Ctrl+C), then a very fast 0.02-second interval producing rapid replies.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ping -i 2 192.168.152.130
PING 192.168.152.130 (192.168.152.130) 56(84) bytes of data.
64 bytes from 192.168.152.130: icmp_seq=1 ttl=64 time=0.028 ms
64 bytes from 192.168.152.130: icmp_seq=2 ttl=64 time=0.073 ms
64 bytes from 192.168.152.130: icmp_seq=3 ttl=64 time=0.054 ms
64 bytes from 192.168.152.130: icmp_seq=4 ttl=64 time=0.067 ms
64 bytes from 192.168.152.130: icmp_seq=5 ttl=64 time=0.058 ms
64 bytes from 192.168.152.130: icmp_seq=6 ttl=64 time=0.072 ms
^C
--- 192.168.152.130 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 10201ms
rtt min/avg/max/mdev = 0.028/0.058/0.073/0.015 ms
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ping -i 0.02 192.168.152.130
PING 192.168.152.130 (192.168.152.130) 56(84) bytes of data.
64 bytes from 192.168.152.130: icmp_seq=1 ttl=64 time=0.038 ms
64 bytes from 192.168.152.130: icmp_seq=2 ttl=64 time=0.028 ms
64 bytes from 192.168.152.130: icmp_seq=3 ttl=64 time=0.052 ms
64 bytes from 192.168.152.130: icmp_seq=4 ttl=64 time=0.038 ms
64 bytes from 192.168.152.130: icmp_seq=5 ttl=64 time=0.036 ms
64 bytes from 192.168.152.130: icmp_seq=6 ttl=64 time=0.058 ms
64 bytes from 192.168.152.130: icmp_seq=7 ttl=64 time=0.032 ms
64 bytes from 192.168.152.130: icmp_seq=8 ttl=64 time=0.046 ms
64 bytes from 192.168.152.130: icmp_seq=9 ttl=64 time=0.029 ms
64 bytes from 192.168.152.130: icmp_seq=10 ttl=64 time=0.034 ms
64 bytes from 192.168.152.130: icmp_seq=11 ttl=64 time=0.045 ms
64 bytes from 192.168.152.130: icmp_seq=12 ttl=64 time=0.037 ms
64 bytes from 192.168.152.130: icmp_seq=13 ttl=64 time=0.038 ms
64 bytes from 192.168.152.130: icmp_seq=14 ttl=64 time=0.025 ms
64 bytes from 192.168.152.130: icmp_seq=15 ttl=64 time=0.029 ms
```

22. Ping with Custom Packet Size & DNS Failure (ping -s 36 and ping -a www.google.com)

The -s flag sets a custom ICMP payload size (36 bytes, so total packet size is 44 bytes).

The next command tries to ping www.google.com by hostname but fails with "Temporary failure in name resolution," since the VM has no working DNS/internet access.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ping -s 36 192.168.152.130
PING 192.168.152.130 (192.168.152.130) 36(64) bytes of data.
44 bytes from 192.168.152.130: icmp_seq=1 ttl=64 time=0.022 ms
44 bytes from 192.168.152.130: icmp_seq=2 ttl=64 time=0.075 ms
44 bytes from 192.168.152.130: icmp_seq=3 ttl=64 time=0.081 ms
44 bytes from 192.168.152.130: icmp_seq=4 ttl=64 time=0.048 ms
44 bytes from 192.168.152.130: icmp_seq=5 ttl=64 time=0.082 ms
44 bytes from 192.168.152.130: icmp_seq=6 ttl=64 time=0.060 ms
44 bytes from 192.168.152.130: icmp_seq=7 ttl=64 time=0.046 ms
44 bytes from 192.168.152.130: icmp_seq=8 ttl=64 time=0.047 ms
44 bytes from 192.168.152.130: icmp_seq=9 ttl=64 time=0.083 ms
^C
--- 192.168.152.130 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8202ms
rtt min/avg/max/mdev = 0.022/0.060/0.083/0.020 ms
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ping -a www.google.com
ping: www.google.com: Temporary failure in name resolution
```

23. Adding a Static ARP Entry (arp -s) and Viewing the ARP Table (arp -a)

Manually maps IP 192.168.152.50 to a MAC address in the ARP cache (marked PERM), then lists all known IP-to-MAC mappings on the subnet, including incomplete (unresolved) entries.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# arp -s 192.168.152.50 00:0c:29:77:33:a5
root@siesgst-VMware-Virtual-Platform:/home/siesgst# arp -a
? (192.168.152.50) at 00:0c:29:77:33:a5 [ether] PERM on ens33
? (192.168.152.0) at <incomplete> on ens33
? (192.168.152.5) at <incomplete> on ens33
? (192.168.152.9) at <incomplete> on ens33
```

24. Viewing ARP Table for a Specific Interface (arp -a -i ens33)

Same ARP table view as above, filtered to only show entries learned on the ens33 interface.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# arp -a -i ens33
? (192.168.152.50) at 00:0c:29:77:33:a5 [ether] PERM on ens33
? (192.168.152.0) at <incomplete> on ens33
? (192.168.152.5) at <incomplete> on ens33
? (192.168.152.9) at <incomplete> on ens33
```

25. Final Interface Check (ifconfig ens33)

A final look at ens33, confirming the IP configuration is stable and showing cumulative RX/TX packet counters after all the preceding tests.

```
root@siesgst-VMware-Virtual-Platform:/home/siesgst# ifconfig ens33
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.152.130  netmask 255.255.255.0  broadcast 192.168.152.255
        ether 00:0c:29:77:33:a5  txqueuelen 1000  (Ethernet)
        RX packets 1850  bytes 1104442 (1.1 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 635  bytes 89878 (89.8 KB)
        TX errors 0  dropped 1 overruns 0  carrier 0  collisions 0
```

CONCLUSION:

This exercise demonstrated Linux network management across interfaces, routing, and diagnostics. Using ifconfig, network interfaces were dynamically configured with IP addresses, netmasks, and operational states. Subnet traffic control was established by adding, replacing, and deleting static routes via ip route. Finally, connectivity and path mapping were verified using diagnostic utilities like ping, traceroute, netstat, and arp.